C PROGRAMMING FROM SCRATCH · EPISODE 05

printf and Library Functions

Printing with the Standard Library

KEY POINTS

- printf prints the text in double quotes to standard output, usually the screen.
- Placeholders %d, %c and %f print an integer, a character and a floating-point number.
- printf is declared in stdio.h (standard input and output). Without the #include, the build fails.
- The C standard specifies library functions; compiler vendors implement them.

01 What printf Does

▶ VIDEO 00:38

- printf prints the text between the double quotes to **standard output**: the screen.
- Text in double quotes is a **string**. `\n` starts a new line.

02 Numbers and Characters

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A **placeholder** holds a spot in the string; the value after the comma fills it in.

PLACEHOLDER	PRINTS	VALUE	CONSOLE
`%d`	an integer, in decimal form	`100`	`100`
`%c`	a single character	`'X'`	`X`
`%f`	a floating-point number (6 decimal places by default)	`3.14`	`3.140000`

placeholders.c — C SOURCE

```c
#include <stdio.h>

int main()
{
    printf("%d\n", 100);
    printf("%c\n", 'X');
    printf("%f\n", 3.14);
    return 0;
}
```

Microsoft Visual Studio Debug Console

```
100
X
3.140000

...\x64\Debug\first_program.exe exited with code 0.
Press any key to close this window . . .
```

03 Why We Need stdio.h

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Without `#include <stdio.h>` the build fails: printf is undefined, and the linker can't find it.

Output · Show output from: Build

```
1>------ Build started: Project: first_program, Configuration: Debug x64 ------
1>hello.c
1>hello.c(5,5): warning C4013: 'printf' undefined; assuming extern returning int
1>hello.obj : error LNK2019: unresolved external symbol printf referenced in function main
1>first_program.exe : fatal error LNK1120: 1 unresolved externals
========== Build: 0 succeeded, 1 failed, 0 up-to-date, 0 skipped ==========
```

> KEY IDEA **stdio = standard input and output**
>
> printf is declared in stdio.h, so any file that uses printf includes stdio.h at the top.

04 Library Functions

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A **library function** is written once and used by everyone, so nobody rewrites printf. The C standard **describes** the functions; compiler vendors **implement** them.

WHO	DOES WHAT
The C standard	specifies each function's **name**, **what it does**, its **parameters** and **return value** (e.g. `printf`, `scanf`, `strlen`)
Compiler vendors	write the code: **Microsoft** (MSVC, Visual Studio, Windows), **Apple** (Clang, Xcode, Mac)
Together	the **C standard library**
Vendor extensions	extra functions some vendors add; not standard C, may not work elsewhere

05 Finding Library Functions

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Each library function is declared in a **header file**; include that header to use it.

HEADER	COVERS	EXAMPLES
`stdio.h`	input and output	`printf`, `scanf`
`math.h`	math functions	`sqrt`, which computes a square root

To look one up: **cplusplus.com** → C library → a header (e.g. stdio.h) → **Functions** → click a name.

CODE 4 files on GitHub

github.com/abitclearer/c-programming-from-scratch/tree/main/05-printf-and-library-functions